

Why is water usage such a big issue with AI? Data centers are usually water cooled and have been for decades.

Posted by u/two_three_five_eighth • • 0 points • 17 comments

I'm in IT and several people have asked me why AI uses so much water. I've also seen several news stories talking about AI water usage.

The thing is, they're always talking about water usage at a data center which would likely also handle streaming, telecom, indexing, and several other task.

AI is not the only thing that can be heavily optimized to run on GPUs.

And every data center I've ever seen uses a closed water loop, so the water is recycled and never released back to the atmosphere.

What made people suddenly care about data center water usage?

Comments

u/Steakbake01 • 1 points • 2026-03-09 14:00 UTC

The rise of AI is leading to more data centers than before, and some of them are making people sick, like Grok's one in Memphis, which is polluting the water people drink

u/AwarenessNo4986 • 1 points • 2026-03-09 08:21 UTC

- 1) not all are closed loop. Correct me if I'm wrong
- 2) the scale at which it is growing
- 3) the overall impact.

I personally can't care less, but that's probably the reasoning

u/ExhaustedByStupidity • 1 points • 2026-03-09 05:21 UTC

It's about quantity.

We're building enough AI datacenters that all the memory that will be made in the next two years has already been purchased. Want to order hard drives from the manufacturer? You're getting them late next year. Need to find components? You're going to have to pay something like 5x what you would have paid six months ago.

We're just building huge AI datacenters as fast as we can. It's much harder to accommodate rapid growth than slow growth.

u/MukadeYada • -2 points • 2026-03-09 04:28 UTC

People are looking for a reason to hate AI.

Right now, on Earth, roughly 50% of humans poop into clean drinking water. There are a very few countries, like Hong Kong, that use grey water (like post-shower water) for their toilets. But in most "modern" nations, we fill our toilets with nearly a gallon of fresh, drinkable water, and then we poop into it, and then we flush it into the sewer.

In the United States, the entire data center industry consumes 785 billion gallons of water a year. "Consumes" doesn't mean "eliminates," since it's all still part of the water cycle. It just means it takes available water out of the system and temporarily puts it back into the water cycle until it eventually falls down again as rain or snow.

But toilets? They use 135 trillion gallons of water a year. That's 135,000 billion - 172 times more than data centers.

If we're really worried about how much water we're denaturing (not "using up," because, again, it all falls down as rain and snow eventually) then there's no reason to freak out about data centers. Our focus should be on toilets.

For someone to worry about the 0.1 gallon of water their friend used making AI pictures, and then go excuse themselves to poop into 1.0 gallons of water - well, it's just Luddism, plain and simple. They're looking for an excuse to hate all things AI, and they found one.

u/panaceaXgrace • 0 points • 2026-03-09 04:23 UTC

The Henderson NV center was built five years ago. It holds a couple hundred petabytes on 64 acres.

The one in my backyard here in Memphis that just opened holds over a megabyte. That's a huge difference in the size, the energy and the water. It's on 522 acres.

The one in Memphis was supposed to use 90% gray water but this never happened. They are not at this time processing water in a closed loop either although they do have the building I believe and they're working on it over a year later. It's taking an extraordinary amount of gas and water right now and will still use water, but the plant will use even more energy. There are so many more planned, so we should be concerned especially when the companies building them are deceiving people like Musk did here. I'm sure he's not alone, I'm just witnessing this firsthand.

u/hikeonpast • 3 points • 2026-03-09 04:41 UTC

You might want to check your units. 200 petabytes is much bigger than a megabyte. Both are a measure of data storage, not power consumption nor power density per rack.

u/panaceaXgrace • 1 points • 2026-03-10 04:35 UTC

I honestly meant exabite which is what Musk said xAI would have the capacity to contain. Whether it does or not is a mystery at this point. Surely you could figure that out though.

I do know what this stuff is, it was a mistake. I am familiar with what a megabyte is. I work in cyber security. I trained AI for years. But I am human. :)

u/archpawm • 1 points • 2026-03-09 10:01 UTC

They said *over* a megabyte. /r/technicallythetruth

u/panaceaXgrace • 1 points • 2026-03-10 04:36 UTC

Thank you. I'm going to call myself vindicated.

But that was a pretty dumb mistake on my part.

u/Delehal • 2 points • 2026-03-09 04:19 UTC

The weird thing is that data centers account for well under 1% of total water usage in the US. The vast majority of water usage is for other industrial and agricultural uses. The amount used for data centers is less than we use for watering the grass on golf courses, or on residential lawns.

Water usage by AI products has also improved quite a bit over time. Some of the most doom-filled predictions are based on years-old data using less efficient systems than the ones that are in use today.

I'm not saying there isn't a problem. I'm just saying that pop culture's perception of the problem seems to have become a bit warped. People *feel* like it's true, so they don't bother to check if it actually is or not.

Now, there are still some serious issues. Probably the biggest problem is that a lot of data centers are built under permits that are managed at city/town level, but they can have an impact much bigger than that. These are really major infrastructure projects that can impact an entire area such as a county or state, but they aren't being planned or managed at that level. As a result, there are many places where data centers have moved in and started using up tons of utilities, which has caused shortages and rising costs for everyone in those areas. Could this be mitigated with better planning? Probably a lot of it could be, yes, but that isn't what people are asking for.

u/archpawm • 1 points • 2026-03-09 04:16 UTC

[It isn't.](<https://andymasley.substack.com/p/a-cheat-sheet-for-conversations-about?open=false#%C2%A7water>)

It can be a problem if people build a new data center in an area not designed to pipe that much water to that spot, but in terms of using up limited resources, AI, especially if you're only looking at chatbots, uses a tiny fraction of the water that a lot of things nobody cares about like paper use.

The reason people make a big deal out of it is that they're against AI for other reasons, and also they often have no understanding of how much water people consume in general. And often compare the water usage of an entire datacenter or training a model to what one person uses.

> And every data center I've ever seen uses a closed water loop, so the water is recycled and never released back to the atmosphere.

Most of the water is used by the power plants generating electricity rather than the datacenters directly.

u/Kakamile • 1 points • 2026-03-09 09:16 UTC

That looks like classic propaganda.

Large wall of text but only one ai stat with no source. It's also claiming per prompt and not the continuous training.

u/archpawm • 1 points • 2026-03-09 09:43 UTC

> Large wall of text but only one ai stat with no source.

Googling around, [here's the source.]

(<https://cloud.google.com/blog/products/infrastructure/measuring-the-environmental-impact-of-ai-inference/>) You'll find many more links littered throughout the article.

If you can find an equally good source saying a day's use of AI uses water comparable to a cheeseburger, I'd be very interested.

The large wall of text is because it's trying to thoroughly address all the incredibly misleading arguments people keep making. If you read them, it makes sense.

> It's also claiming per prompt and not the continuous training.

[This section](<https://andymasley.substack.com/p/a-cheat-sheet-for-conversations-about?open=false#%C2%A7training-an-ai-model-uses-too-much-energy>) talks about that, with a source. Training uses about 40% the energy (and by extension water) as general use. So just add 40% to negligible and it's still negligible.

I'm not going to blame you for not reading the whole thing. Like you said, it's a large wall of text. Feel free to ask more questions and I'll link you to the appropriate section.

Though I do recommend at least looking through the [contents]

(<https://andymasley.substack.com/p/a-cheat-sheet-for-conversations-about?open=false#%C2%A7contents>).

u/Kakamile • 1 points • 2026-03-09 17:15 UTC

I mean that was entirely a large wall of text just for the water section, despite the entire water section being just one ai claimed data point, which is only the claim for per prompt. They fluffed it up with playing examples built on assuming that one claim.

Previous conversations by analysts already show how you can show ai water use as low or high depending on which you measure, which is why they use both.

So this blog using just one and inflating it for pages is a problem.

Plus, this denial is despite videos of people who live by ai data centers getting less water, having audio and light noise, of local energy grids showing large frequency noise due to the data centers, and news and medical reports on increases in sickness due to subaudible tones.

u/archpawn • 1 points • 2026-03-09 19:32 UTC

> Previous conversations by analysts already show how you can show ai water use as low or high depending on which you measure, which is why they use both.

Googling around, I found a source saying [519 milliliters] (<https://www.washingtonpost.com/technology/2024/09/18/energy-ai-use-electricity-water-data-centers/>). Small enough that you'd need 300 prompts a day to equal 1% an average person's water consumption. Very different from the one

they linked to, but still very low.

Can you show me an example saying it's high?

> Plus, this denial is despite videos of people who live by ai data centers getting less water,

Opening a data center that uses a ton of water all in one place can cause problems for grids that aren't used to it. And they do a lot more than just chatbots.

u/WorldTallestEngineer • 3 points • 2026-03-09 04:09 UTC

One thing that makes AI data centers different is the intense energy density. Yeah I data centers are using up to one megawatt of power per server rack.

So the cooling requirements are a lot more intense. And this can mean evaporating water directly to atmosphere in order getting off cooling capacity.

u/Significant-Catch174 • 6 points • 2026-03-09 04:08 UTC

The amount of data centers is increasing dramatically and people are learning more about their consumption.